

Grasp-TAGS: Task-Aware Grasp Selection via Critic Distillation for Prehensile Loco-Manipulation

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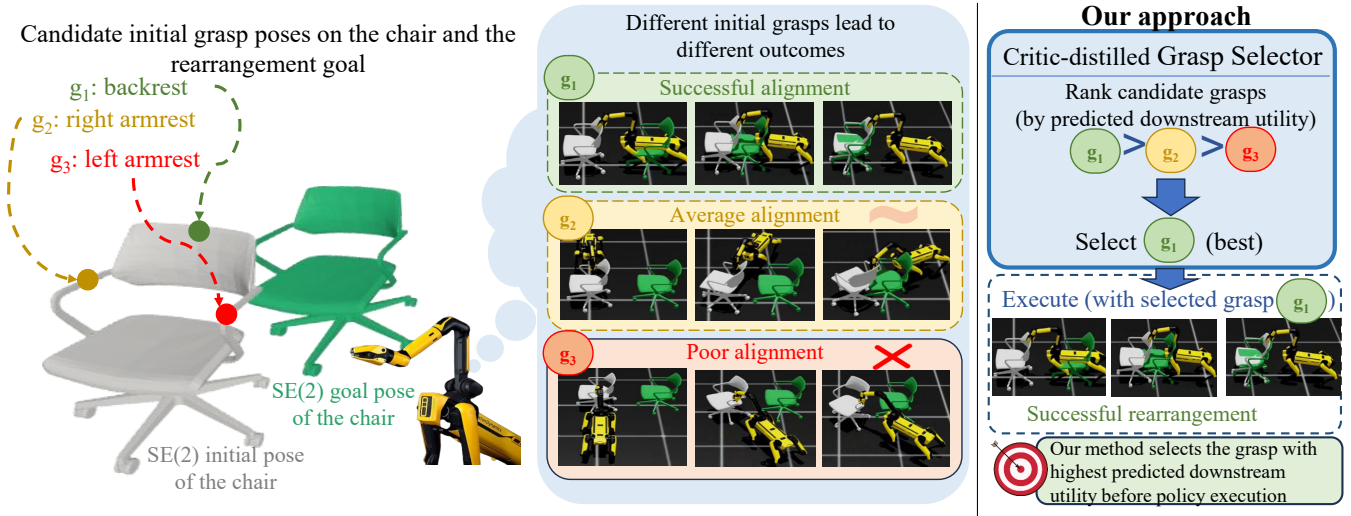


Fig. 1: Overview of the proposed framework Grasp-TAGS. Different feasible grasp poses on the same object can induce substantially different downstream loco-manipulation behaviors and rearrangement outcomes. We therefore train a manipulation policy conditioned on multiple initial grasp configurations via PPO and distill the learned critic into a lightweight grasp selection network. At inference time, the selector ranks candidate grasps according to their predicted downstream task utility, selects the most suitable grasp for the desired object pose, and executes the manipulation policy from the selected grasp configuration.

Abstract—Loco-manipulation, in which mobile manipulators coordinate locomotion and manipulation over long horizons, has become an important paradigm for large-object rearrangement in applications such as service robotics and warehouse automation. Despite recent progress, reliably executing these tasks remains challenging because performance depends on complex robot-object interaction dynamics. Existing approaches address this challenge either through model-based methods, which often require accurate dynamics models, or through learning-based methods, which reduce this requirement but typically assume a fixed or predefined initial grasp. However, different feasible grasps can lead to substantially different downstream manipulation performance. Therefore, we propose Grasp-TAGS, a task-aware grasp selection framework for goal-conditioned large-object rearrangement. We train a unified PPO policy across multiple initial grasp configurations and distill the learned critic into a lightweight grasp utility network that ranks candidate grasps using only pre-interaction grasp and goal information. At inference time, we select the grasp with the highest predicted downstream utility for the downstream policy to complete the task. Experiments on household objects in simulation show that Grasp-TAGS effectively improves rearrangement success over both fixed single-grasp policies and random grasp selection and reduces final orientation error across all evaluated objects. For more qualitative results, refer to the supplementary video and webpage <https://rpm-lab-umn.github.io/superq-grasp-webpage/>.

I. INTRODUCTION

Loco-manipulation enables a mobile robot to coordinate base locomotion and manipulation to physically interact with objects and accomplish manipulation tasks. Compared with traditional fixed-base manipulation, it is particularly well suited to large-object rearrangement, where bulky or heavy objects beyond conventional tabletop settings, such as chairs, carts, and luggage, must be transported to desired poses [1]–[4]. In prehensile large-object rearrangement, the robot maintains sustained gripper-object contact while the object remains grounded during manipulation. The large workspace and complex interaction dynamics of these tasks require coordinated base and arm motion, often exceeding the workspace and capabilities of a fixed-base manipulator. Developing robust loco-manipulation skills for such tasks is therefore important for applications including service robotics, warehouse automation, and assistive robotics.

Traditionally, loco-manipulation skills for large-object rearrangement have been addressed using model-based approaches such as model predictive control (MPC), by explicitly modeling system dynamics and generating whole-body motions through analytical planning and control [5], [6]. These approaches generally rely on sufficiently accurate knowledge of the underlying dynamics. More recently, learning-based methods using reinforcement learning and imitation learning have demonstrated promising performance

[4], [7]–[9]. By learning control policies from simulated interactions or demonstrations, these methods reduce the need to explicitly model complex robot-object dynamics during execution. However, existing learning-based approaches typically assume a fixed or predefined initial grasp configuration before downstream manipulation.

However, the choice of initial grasp can substantially influence the subsequent rearrangement process. Beyond grasp stability, different grasp configurations induce different arm configurations, contact locations, and manipulation dynamics during pushing or dragging, which can directly affect task controllability and success [10], [11]. Consequently, even among feasible and stable grasps, downstream performance can vary substantially depending on the rearrangement goal. This suggests that large-object rearrangement should not rely on a fixed or predefined initial grasp. Moreover, grasp selection in the large-object rearrangement should account not only for grasp feasibility, but also for the expected utility of each grasp for the downstream manipulation task.

To this end, we propose Grasp-TAGS, a Task-Aware Grasp Selection framework for goal-conditioned large-object rearrangement, as illustrated in Fig. 1. Grasp-TAGS evaluates candidate grasps according to their expected downstream task utility and selects the most suitable grasp before manipulation begins. We first train a unified manipulation policy via PPO across multiple feasible initial grasp configurations, allowing both the policy and its critic to capture grasp-dependent downstream behavior. Consequently, the critic assigns different value estimates to initial states associated with different candidate grasps. We then distill these grasp-dependent values into a lightweight grasp utility network using only information available prior to physical interaction. At inference time, the network ranks candidate grasps for the specified rearrangement goal, selects the grasp with the highest predicted utility, and executes the unified manipulation policy from the selected configuration. We evaluate the proposed framework on goal-conditioned rearrangement of multiple large household objects in simulation to validate the pipeline’s effectiveness in selecting the optimal grasp configuration for large-object rearrangement.

In summary, our contributions are:

- We develop a unified goal- and grasp-conditioned loco-manipulation policy trained via PPO that operates across multiple initial grasp configurations for large-object rearrangement.
- We distill the critic into a lightweight grasp utility network that predicts downstream grasp utility from pre-interaction grasp and goal information, enabling task-aware grasp selection at inference time.
- We conduct preliminary simulation experiments showing that different feasible grasps can lead to substantially different downstream manipulation outcomes, and that our proposed grasp selection framework, Grasp-TAGS, improves task success over fixed or random grasp selection.

II. RELATED WORK

A. Large Object Rearrangement in Loco-Manipulation

Large-object rearrangement remains a challenging problem in Loco-manipulation. In this work, we refer to large objects as bulky objects that extend beyond the conventional tabletop manipulation setting, such as chairs and boxes [1]. Their size and weight require robots to coordinate base locomotion and end-effector motion to transport the object toward a desired pose [2]–[4].

Early approaches primarily relied on model-based methods, such as model-predictive control (MPC), to explicitly account for object dynamics and execute whole-body motions analytically in accordance with traditional physical laws [5], [6], [12]–[14]. More recently, data-driven approaches based on reinforcement learning (RL) and imitation learning (IL) have emerged as an alternative to directly handling the complex interaction dynamics of large-object manipulation by learning the knowledge of interaction [3], [4], [7]–[9], [15]–[17]. For instance, Ravan et al. [8] develop a teleoperation-based pipeline to collect demonstrations of a quadrupedal robot and subsequently train a diffusion policy [18]. In contrast, ALORE [7], RobotMover [4], and ReLIC [9] adopt hierarchical reinforcement learning to learn manipulation policies in simulation and transfer them to real-world platforms. Such approaches reduce the reliance on costly expert demonstrations and enable scalable policy learning through simulation.

Despite these advances, existing learning-based approaches [3], [4], [7], [8] typically assume a single predefined initial grasp pose (with positional variance smaller than 10 cm and on the same affordance part of the object). This assumption overlooks that larger-scale grasp configurations (positional variance ≥ 30 cm) on different affordance parts of the object can induce substantially different interaction dynamics, leading to different downstream manipulation performance. Some works [14], [19] partially address this issue by dynamically sampling multiple contact points during RL training, but primarily consider geometrically simple bulky objects. In contrast, we treat grasp pose as a decision variable in goal-conditioned large-object rearrangement and explicitly select the grasp configuration most suitable for downstream prehensile manipulation of geometrically complex household objects, such as chairs, toward desired poses.

B. Grasp Planning and Task-Oriented Grasp Selection

Grasp planning aims to identify end-effector poses that establish stable contact with a target object [1]. Given that multiple feasible grasps may exist, grasp pose evaluation and selection determine which candidate to execute. Traditional approaches evaluate grasps using analytical criteria such as contact forces, object stability, and wrench-based measures [1], [20]–[22]. More recent learning-based methods estimate grasp quality directly from visual observations or point clouds [23]–[25] using some expert datasets.

These approaches primarily assess the feasibility and stability of the grasp itself, without considering its impact

on downstream manipulation. Recent task-oriented grasping methods address this limitation by selecting grasps according to the intended manipulation objective [11], [26]–[29]. However, existing studies predominantly focus on tabletop manipulation, where tasks are relatively short-horizon, and objects are typically smaller and geometrically simpler. In contrast, large-object rearrangement involves long-horizon loco-manipulation of geometrically complex objects [1], [8], where grasp choice can substantially affect downstream task execution. In this work, we investigate grasp pose evaluation and selection for goal-conditioned large-object rearrangement, evaluating candidate grasps according to their expected utility for the downstream loco-manipulation task.

C. Conditioned Policy Learning

Conditioned policy learning incorporates task-level or contextual variables into policy learning, enabling a single policy to adapt its behavior to different task objectives and environments. In imitation learning, policies have been conditioned on variables such as target object poses [18], [30], [31], language instructions [32]–[35], and task parameters [36]. Reinforcement learning similarly enables policies to condition on task parameters [4], [7], [37], object physical attributes [38], and language instructions [39]–[42], allowing agents to acquire diverse behaviors through self-exploration rather than expert demonstrations [43]–[47].

Despite the success of conditioned policy learning, initial grasp pose has received limited attention as a conditioning variable for large-object loco-manipulation. A few recent works incorporate grasp pose into imitation-learning-based policies [10], [48]–[50], but rely on expert demonstrations to learn the corresponding manipulation behaviors. The work most closely related to ours [11] incorporates grasp information into value-based grasp evaluation, but focuses on small, geometrically simple objects in tabletop settings. In contrast, we explicitly condition an RL-based loco-manipulation policy on grasp pose, enabling it to capture diverse interaction dynamics. On this policy, we further distill its task-level value into a grasp-selection agent that estimates the expected downstream utility of candidate grasps before policy execution.

III. PROBLEM STATEMENT

We consider goal-conditioned large-object rearrangement, where a mobile manipulator moves a target object from an initial pose p_0 to a desired planar pose $p_{goal} = (x, y, \theta) \in SE(2)$, given a set of feasible initial grasp poses

$$\mathcal{G} = \{g_1, g_2, \dots, g_K\}, \quad (1)$$

In a large-object rearrangement task, different grasps may induce different downstream manipulation performance for the same rearrangement goal.

Our objective is twofold. First, we learn a manipulation policy $\pi(a_t|o_t, g_i)$ capable of executing the rearrangement task under different $g_i \in \mathcal{G}$. Second, we select the grasp providing the highest expected downstream task utility:

$$g^* = \arg \max_{g_i \in \mathcal{G}} U(g_i, p_0, p_{goal}), \quad (2)$$

where $U(\cdot)$ represents the expected utility of executing the learned manipulation policy from grasp g_i . The selected grasp g^* is used for downstream manipulation at inference time.

IV. METHODOLOGY

Unlike conventional mobile manipulation settings that assume a fixed grasp configuration, we consider multiple feasible grasp poses on the target object. Each grasp pose induces a distinct robot–object interaction configuration and, consequently, different manipulation dynamics. Our objective is two-fold: (1) learn a unified manipulation policy capable of transporting the object to a desired pose under different grasp configurations, and (2) select, at inference time, the grasp pose with the highest expected utility for successfully completing the rearrangement task (see Figure .2).

A. Hierarchical Policy Learning

Large-object rearrangement requires coordinated locomotion and manipulation over long horizons [2]. Directly learning a unified whole-body policy over all robot joints is challenging due to the high-dimensional action space and complex robot-object interactions [4], [7]. To improve learning efficiency, inspired by [9], [51], we adopt a hierarchical control architecture consisting of a pretrained low-level locomotion controller and a high-level task policy for manipulation. The low-level controller maintains locomotion stability, while the high-level policy generates task-related commands for object rearrangement. This decomposition allows the reinforcement learning agent to focus on task-level manipulation and planning without explicitly learning low-level locomotion dynamics (see Fig. 2)

1) *Low-level controller*: The low-level controller generates dynamically stable locomotion behaviors while tracking task-specific commands from the high-level controller. Specifically, the high-level command is defined as

$$a_t^{high} = (v_{base,x}, v_{base,y}, \omega_{base,z}, \Delta q_{1:6}) \quad (3)$$

where $v_{base,x}, v_{base,y}$ are the base linear velocities in the robot frame, $\omega_{base,z}$ is the base yaw rate in the robot frame, and $\Delta q_{1:6}$ are the arm joint angle displacements. Inspired by [9], the 6-dim arm commands are executed directly with PD controllers to hold the object, and base velocity commands are processed with a pretrained locomotion policy to maintain stability and track the velocity. The locomotion policy receives the observation

$$o_t^{low} = (v_{base,t}, g_t, q_{1:18}, \dot{q}_{1:18}, a_{t-1}^{low})$$

where $s_{base,t} \in \mathbb{R}^6$ denotes the observed base state, $g_t \in \mathbb{R}^3$ denotes the projected gravity vector, and $q_{1:18}, \dot{q}_{1:18} \in \mathbb{R}^{18}$ denote the joint positions and velocities of all robot joints, respectively. $a_t^{high} \in \mathbb{R}^9$ denotes the high-level command defined in Eq. 3. The locomotion policy produces 12-dimensional leg joint commands $a_{low,t} = q_{leg,0:12} \in \mathbb{R}^{12}$, which are executed by the robot to maintain locomotion stability and track the desired base velocity.

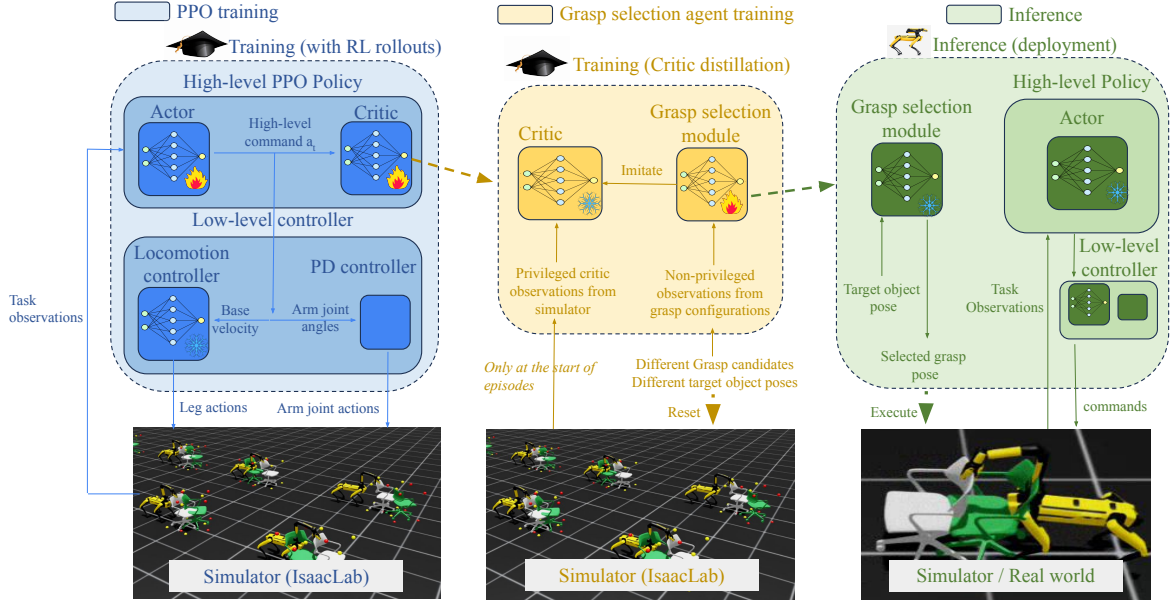


Fig. 2: **Overview of the proposed pipeline Grasp-TAGS.** A hierarchical policy trained via PPO performs loco-manipulation using a pretrained low-level controller. The trained critic is subsequently distilled into a grasp selection module, which selects the grasp g^* with the highest predicted downstream utility for the target object pose before policy execution.

TABLE I: Observation terms to the high-level controller. It consists of proprioceptive data from the robot and the real-time observations of the object.

Obs. Terms	Dim	Description
$q_{1:18}$	18	Position of all the joints (both arm & legs)
$\dot{q}_{1:18}$	18	Velocity of all the joints
$s_{base,t}$	2	Orientation of the base (no yaw)
$\dot{s}_{base,t}$	3	Angular velocity of the base
a_{t-1}^{high}	9	Last action
$ee_{base,t}$	7	SE(3) pose of the end-effector in robot frame
$obj_{base,t}$	3	SE(2) pose of the object in robot frame
$goal_{obj,t}$	3	Target SE(2) pose in object frame

2) *High-level controller*: Built upon the pretrained locomotion controller, the high-level controller executes a policy focused on task-level planning and manipulation. Following [7], the observations provided to the high-level controller are summarized in Table I. The output of the high-level controller is the high-level command defined in Eq. 3, which is subsequently provided to the low-level controller for execution.

We train the high-level controller using Proximal Policy Optimization (PPO) implemented in RSL-RL [52], while keeping the pretrained low-level locomotion policy fixed throughout training. The objective is to maximize the expected discounted cumulative reward:

$$J(\pi_{\theta,high}, \pi_{low}) = \mathbb{E}_{r \sim p(\cdot | \pi_{high}, \pi_{low})} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]$$

where the reward r_t is composed of the terms described in Sec. IV-C. More training details can be found in Sec. IV-B

B. Grasp-Conditioned Loco-manipulation Policy Training

Rather than learning an independent high-level policy for each grasp, we train a grasp-conditioned policy via PPO [47] that operates across multiple initial grasp configurations, eliminating the need for separate training for each grasp on the same object. Following [7], the initial grasp pose is implicitly encoded by the arm joint angles.

During training, each episode samples one of several predefined feasible grasp poses on the target object. After establishing the grasp, the robot is required to transport the object to a desired target pose (x, y, θ) , where $x, y \in [-1, 1]$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. The policy therefore learns manipulation behaviors across different initial grasp configurations and rearrangement objectives. We implement the training environment in IsaacLab [53] and train the high-level controller using PPO with 4096 parallel environments for 8000 iterations, as described in Sec. IV-A.2.

Training a unified policy across multiple grasp poses provides two key advantages. First, it enables the policy to capture the diverse interaction dynamics induced by different robot-object contact configurations without requiring separate training procedures for each grasp pose. Second, the resulting value function provides a common measure of the expected long-horizon manipulation performance across candidate grasp poses, which serves as the foundation for subsequent grasp pose selection.

C. Task-Specific Reward Functions

We design the reward function for the high-level controller according to three objectives: (i) task completion, which encourages the robot to transport the object to the de-

sired pose; (ii) motion smoothness, which promotes smooth and non-aggressive manipulation behaviors; and (iii) safety, which penalizes undesired robot-object contacts and object toppling. The complete set of reward terms is summarized in Table II.

D. Critic Distillation for Task-Aware Grasp Selection

As illustrated in Fig. 2, the PPO critic [47] estimates the expected cumulative return from the current critic observation. Thus, the critic implicitly captures the expected downstream utility of different grasp configurations: grasps that enable more successful, efficient, and stable rearrangement should receive higher value estimates. At the beginning of an episode, different initial grasp poses $g_1, g_2, \dots$ induce different critic observations $o_0^c(g_1), o_0^c(g_2), \dots$, and consequently different value estimates

$$V^\pi(o_0^c(g_i)) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t \mid o_0^c(g_i) \right]. \quad (4)$$

However, the critic relies on privileged information unavailable at inference time, including object physical properties such as mass and velocity. Although these properties can potentially be estimated through physical interaction [7], [38], such interaction may require committing to an initial grasp. If that grasp is suboptimal, the robot must release and re-grasp the object, introducing additional manipulation steps and potential failure modes. We therefore avoid explicit physics estimation and instead distill the critic's grasp-dependent value into a lightweight grasp selection network $V_{\text{grasp}}(x_i)$ into a grasp utility network using only pre-interaction information:

$$x_i = \left(g_i, p_{\text{goal}}^{\text{obj}} \right), \quad (5)$$

where g_i represents the candidate grasp configuration, and $p_{\text{goal}}^{\text{obj}}$ is the target pose represented in the initial object frame, representing the task objective. By removing privileged physical variables from the input, the grasp selection network learns the expected grasp utility marginalized over the physical-property distribution encountered during training.

After policy training, we sample initial states from all N sub-environments and regress the utility network toward the corresponding critic values:

$$\mathcal{L}_{\text{distill}} = \frac{1}{N} \sum_{k=1}^N \left[V_{\text{grasp}}(x_i^{(k)}) - V^\pi(o_0^{c,k}(g_i^{(k)})) \right]^2. \quad (6)$$

where $g_i^{(k)}$ is the grasp candidate sampled in the k -th sub-environment, $x_i^{(k)}$ is the corresponding pre-interaction information and $o_0^{c,k}(g_i^{(k)})$ is the associated critic observation.

At inference, for the desired target object pose represented as $p_{\text{goal}}^{\text{obj}}$, we select

$$g^* = \arg \max_{g_i \in \mathcal{G}} V_{\text{grasp}}(x_i), \quad (7)$$

The manipulation policy is then executed conditioned on the selected grasp pose g^*

V. EXPERIMENTAL SETUP

We evaluate the proposed grasp-pose-conditioned manipulation policy and critic-distilled grasp pose selection module on large-object rearrangement tasks through quantitative experiments in IsaacLab using synthetic object models.

A. Loco-manipulation Policy Evaluation in Simulation



Fig. 3: **Objects and candidate grasp poses used in the experiments.** The dataset includes three chairs, two carts, and three buckets, with predefined candidate grasps highlighted in red.

Dataset: We construct a dataset of eight common household large objects, including three chairs, two carts, and three buckets, obtained from Fab [54]. These objects represent common household object categories while covering diverse geometric structures, as illustrated in Fig. 3. For each object, we manually select feasible initial grasp configurations and compute their corresponding inverse kinematics solutions. Specifically:

- **Chairs:** three grasp poses on the two armrests and the backrest.
- **Carts:** three grasp poses on the two sides of the basket and the handle.
- **Buckets:** two grasp poses on the two edges.

Large Object Rearrangement Task As described in Sec. III, we implement the large-object rearrangement task in IsaacLab [53]. At the beginning of each episode, the target object is placed at the origin and a target $SE(2)$ pose (x, y, θ) is randomly sampled, where $x, y \in [-1, 1]$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. We then select an initial grasp pose either using the proposed grasp pose selection module or the analytical methods used in the baseline approaches. The robot subsequently executes the selected grasp and the corresponding manipulation policy to transport the object toward the target pose. We evaluate task performance using the metrics described below. All experiments are conducted with 4096 parallel environments. **Evaluation metrics:** We evaluate the performance of the loco-manipulation policy under a given grasp pose using four metrics after a fixed execution horizon of 750 steps:

- 1) **Success rate:** the fraction of parallel sub-environments in which the final planar position error is below 0.1 m and the final angular error is below 10° ;
- 2) **Planar positional error:** measuring the final Euclidean distance on the xy -plane between the observed and target object positions;
- 3) **Angular error:** measuring the final absolute yaw difference between the observed and target object orientations.
- 4) **Keypoint Pose Error:** The angular difference between the observed and target object poses estimated from two sets of 2D object keypoints using the Kabsch–Umeyama alignment algorithm [55]. We use the eight corners of the

TABLE II: Rewarding terms to train the high-level controller

Reward Terms	Formula	Weight	Description
Task Completion	$\mathbb{1}[\ p_o - p_o^*\ < \varepsilon_p]$	10.0	Positive reward when the object reaches the target pose
Keypoint Pose matching	$\exp\left(-\frac{\ \mathbf{K}_{goal} - \mathbf{K}_{obj}\ _2}{\sigma^2}\right)$	8.0	Reduce the distance between the current and target object keypoints
Goal-directed Velocity	$\exp\left(\frac{\dot{\mathbf{v}}_o \cdot \mathbf{d}_{goal}}{\sigma^2} - 1\right)$	-1.0	Encourage alignment between the object velocity and the direction toward the goal
Alive	$\mathbb{1}[episode \text{ is alive}]$	5.0	Sparse reward for maintaining a valid and active episode
Linear Velocity Smoothness	$\ v_o\ ^2$	-2.0	Penalty on changes in the object's linear velocity
Angular Velocity Smoothness	$\ \omega_o\ ^2$	-2.0	Penalty on changes in the object's angular velocity
Orientation Stability	$\ g_{o,xy}\ ^2$	-10.0	Encourage the object to maintain a desired upright/flat orientation
Joint Configuration Regularization	$\ q_{1:6} - q_{default,1:6}\ ^2$	-5.0	Penalty based on the deviation of the robot joint positions from their default positions
Undesired Contact Penalty	$\mathbb{1}[\ F_{o,r}\ > \varepsilon_{contact}]$	-7.0	Penalty for undesired contacts during manipulation

object's bounding box as keypoints. This metric provides a more robust estimate of the orientation error than directly comparing the object yaw angles.

Baseline methods: We establish three baselines to evaluate the benefits of grasp-pose-conditioned policy learning and the proposed task-aware grasp pose selection module for large-object rearrangement. Specifically:

- 1) **PPO_i**: Each baseline consists of a policy trained exclusively on a single grasp pose g_i ($i \in \{1, 2, 3\}$ for the three grasp poses on the chair/cart, $i \in \{1, 2\}$ for the two grasp poses on the bucket) using the same training procedure and hyperparameters. For example, PPO_1 is trained from scratch using only grasp pose g_1 , without exposure to the other candidate grasp poses. These baselines evaluate the effect of training a policy across multiple grasp configurations.;
- 2) **PPO_g+Random**: This baseline uses the proposed grasp-pose-conditioned loco-manipulation policy trained on all grasp candidates, but does not employ the grasp pose selection module. During both training and inference, a candidate grasp pose is randomly sampled at the beginning of each episode. This baseline isolates the contribution of task-aware grasp pose selection.
- 3) **Grasp-TAGS (Ours)**: Our complete pipeline consists of the grasp-pose-conditioned loco-manipulation policy together with the critic-distilled grasp pose selection module. At inference time, the selection module evaluates the candidate grasp poses and selects the one with the highest predicted downstream task utility. This configuration represents the proposed method.

VI. RESULTS

A. Loco-manipulation Policy Evaluation in Simulation

Our preliminary results (see Table III) indicate that the initial grasp configuration substantially affects downstream loco-manipulation performance (see Fig. 4). Individually trained PPO_i policies can outperform PPO_g +Random by achieving a higher success rate and lower positional or angular errors, because each policy specializes in a single robot-object interaction configuration. However, the relative performance of different grasps varies across rearrangement goals, motivating explicit task-aware grasp selection.

Combining the unified policy with the proposed grasp selection network improves overall task success (see Fig. 5).

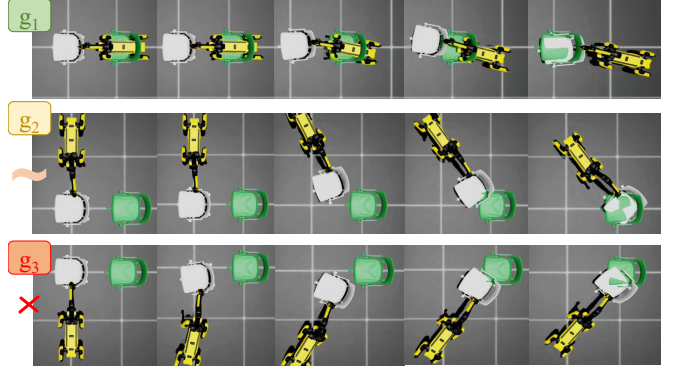


Fig. 4: **Qualitative comparison across initial grasp poses.** The backrest grasp (g_1) yields better final alignment with the target pose than the two armrest grasps (g_2 and g_3).

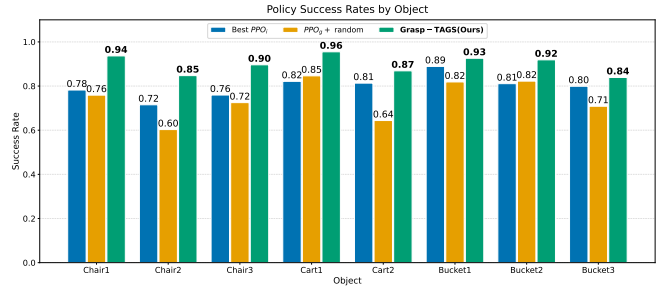


Fig. 5: **Success rates** for the best single-grasp policy, multi-grasp policy + random grasp selection, and Grasp-TAGS. Grasp-TAGS achieves the highest success rate across all evaluated objects.

For Chair1, Grasp-TAGS achieves a success rate of 93.65% over all sub-environments compared with 75.88% for random grasp selection and 69.02–78.20% for the individual single-grasp policies. The proposed method also achieves a lower angular error, suggesting that selecting an appropriate interaction configuration is particularly beneficial for tasks that require coordinated object translation and rotation. The same trend is observed for the other objects.

Observation: The quantitative results lead to the following observations:

- 1) **Success Rate:** Our method achieves a higher success rate than both baselines on most object categories, demonstrating that task-aware grasp pose selection improves the overall probability of successfully completing the target

TABLE III: **Quantitative results on simulated large-object rearrangement.** We compare PPO_i , PPO_g +random, and Grasp-TAGS using success rate, planar positional error, and angular error across multiple objects. Higher success rate and lower errors indicate better performance.

	Chair1			Chair2		
	PPO_i	PPO_g + random	Grasp-TAGS	PPO_i	PPO_g + random	Grasp-TAGS
Success rate	70.09/75.46/78.20	75.88	93.65	63.7971/51/70.00	60.28	84.72
Planar Positional Error	0.0292/0.0250/0.0366	0.0281	0.0162	0.034/0.0161/0.0142	0.045	0.0239
Angular Error	0.3571/0.2850/0.2802	0.2407	0.0712	0.5268/0.3421/0.3732	0.4382	0.1308
Keypoint Angular Error	20.4597/16.3271/16.0567	13.7906	4.0793	30.1858/19.5889/21.3821	25.1091	7.4923
	Chair3			Cart1		
	PPO_i	PPO_g + random	Grasp-TAGS	PPO_i	PPO_g + random	Grasp-TAGS
Success rate	75.93/68.41/72.22	72.49	89.62	82.13/68.48/67.38	84.62	95.51
Planar Positional Error	0.0204 /0.0626/0.0307	0.041	0.0309	0.0291/0.0989/0.0924	0.0426	0.0283
Angular Error	0.3546/0.4425/0.3628	0.3616	0.1212	0.1716/0.2522/0.3781	0.1471	0.0575
Keypoint Angular Error	20.3148/25.3516/20.7884	20.7177	6.9462	9.8312/14.5265/20.9177	8.4009	3.2925
	Cart2			Bucket1		
	PPO_i	PPO_g + random	Grasp-TAGS	PPO_i	PPO_g + random	Grasp-TAGS
Success rate	62.96/81.32/72.75	64.38	86.89	88.89/86.18	81.86	92.58
Planar Positional Error	0.0901/ 0.0235 /0.0762	0.1158	0.0551	0.0101 /0.0280	0.0201	0.0138
Angular Error	0.4028/0.3335/0.3844	0.371	0.1321	0.1428/0.2250	0.2455	0.1279
Keypoint Angular Error	23.0807/19.1110/22.0228	21.2587	7.5734	8.1790/12.8935	14.0673	7.3268
	Bucket2			Bucket3		
	PPO_i	PPO_g + random	Grasp-TAGS	PPO_i	PPO_g + random	Grasp-TAGS
Success rate	76.10/81.08	82.23	91.87	79.91/70.00	70.85	83.89
Planar Positional Error	0.0219/0.0220	0.0144	0.0105	0.0258 /0.0265	0.0689	0.029
Angular Error	0.3370/0.2490	0.2822	0.1343	0.2598/0.3199	0.3609	0.2276
Keypoint Angular Error	19.3076/14.2670	16.1711	7.6976	14.8877/18.3309	20.6764	13.0426



Fig. 6: **Illustrative example of grasp utility predictions for a target object pose (green mesh).** The colors visualize the relative utility estimated by Grasp-TAGS, where green denotes higher predicted downstream utility and red denotes lower predicted utility.

rearrangement task.

- 2) **Planar Positional Error:** The best single-grasp policy PPO_i achieves the lowest mean final planar distance on average. However, our method achieves comparable performance, with a difference of only approximately 3 cm at most from the best-performing baseline. This difference is small relative to the scale of the objects considered in our experiments, which are typically larger than 30 cm.
- 3) **Angular Error and Keypoint Angular Error:** Our method achieves the lowest final angular errors under both metrics, indicating that task-aware grasp selection is particularly effective at aligning the final object orientation with the desired target pose.
- 4) **Effect of Unified Policy Learning:** The single-grasp policies PPO_i generally perform better than the unified policy PPO_g +Random, as expected, since each PPO_i specializes in a single grasp configuration and therefore faces a less diverse training distribution. However, when combined with the proposed grasp pose selection module, the unified policy PPO_g +Selection consistently outperforms the single-grasp baselines in terms of overall

task success. This result demonstrates that the benefit of task-aware grasp selection can outweigh the performance advantage of individually specialized policies.

These results suggest that grasp selection should consider downstream task utility rather than grasp stability alone [20]. In loco-manipulation, the initial grasp influences how effectively the robot controls the object throughout execution. Accordingly, our grasp selection network predicts the utility of candidate grasps and selects the most suitable one for the downstream task (Fig. 6), improving overall rearrangement performance.

VII. CONCLUSION

We presented a task-aware grasp selection framework for prehensile goal-conditioned large-object loco-manipulation. By training one unified manipulation policy across multiple initial grasp configurations, the learned critic captures the long-horizon utility of different grasp configurations. Distilling these values into a lightweight grasp utility network enables the robot to select a suitable grasp before physical interaction without explicitly estimating privileged physical properties of the object. Preliminary simulation results demonstrate improved rearrangement success over fixed and randomly selected grasp configurations. Future work will extend the evaluation to additional household objects and real-world deployment on the SPOT platform.

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REFERENCES

- [1] X. Tu and K. Desingh, "Superq-grasp: Superquadrics-based grasp pose estimation on larger objects for mobile-manipulation," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 3361–3368.
- [2] Z. Fu, X. Cheng, and D. Pathak, "Deep whole-body control: Learning a unified policy for manipulation and locomotion," in *Conference on Robot Learning (CoRL)*, 2022.
- [3] R. Mendonca, E. Panov, B. Bucher, J. Wang, and D. Pathak, "Continuously improving mobile manipulation with autonomous real-world rl," 2024.
- [4] T. Li, J. Truong, J. Yang, A. Clegg, A. Rai, S. Ha, and X. Puig, "Robotmover: Learning to move large objects from human demonstrations," 2025. [Online]. Available: <https://arxiv.org/abs/2502.05271>
- [5] J. Li and Q. Nguyen, "Multi-contact mpc for dynamic loco-manipulation on humanoid robots," *2023 American Control Conference (ACC)*, pp. 1215–1220, 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:252368129>
- [6] X. Wang, Y. Liu, X. Song, Y. Liu, S. Zhang, and S. Jiang, "An interactive navigation method with effect-oriented affordance," *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16 446–16 456, 2024. [Online]. Available: <https://api.semanticscholar.org/CorpusID:272723315>
- [7] Z. Bi, Y. Zhang, K. Chen, G. Zhao, Y. Li, and J. Ma, "Alore: Autonomous large-object rearrangement with a legged manipulator," 2026. [Online]. Available: <https://arxiv.org/abs/2602.04214>
- [8] Y. Ravan, Z. Yang, T. Chen, T. Lozano-Pérez, and L. P. Kaelbling, "Combining planning and diffusion for mobility with unknown dynamics," 2024. [Online]. Available: <https://arxiv.org/abs/2410.06911>
- [9] X. Zhu, Y. Chen, L. Sun, F. Niroui, S. L. Cleac'h, J. Wang, and K. Fang, "Versatile loco-manipulation through flexible interlimb coordination," in *9th Annual Conference on Robot Learning*, 2025. [Online]. Available: <https://openreview.net/forum?id=Spg25qkV81>
- [10] K. Fang, Y. Zhu, A. Garg, A. Kuryenkov, V. Mehta, L. Fei-Fei, and S. Savarese, "Learning task-oriented grasping for tool manipulation from simulated self-supervision," *Robotics: Science and Systems (RSS)*, 2018.
- [11] L. Röstel, D. Winkelbauer, L. Pitz, Johannes Sievers, and B. Bäuml, "Composing dextrous grasping and in-hand manipulation via scoring with a reinforcement learning critic," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, May 2025.
- [12] C. Dai, X. Liu, K. Sreenath, Z. Li, and R. Hollis, "Interactive navigation with adaptive non-prehensile mobile manipulation," 2024. [Online]. Available: <https://arxiv.org/abs/2410.13418>
- [13] K. Ellis, H. Zhang, D. Stoyanov, and D. Kanoulas, "Navigation among movable obstacles with object localization using photorealistic simulation," 07 2022.
- [14] M. Polverini, A. Laurenzi, E. Mingo Hoffman, F. Ruscelli, and N. Tsagarakis, "Multi-contact heavy object pushing with a centaur-type humanoid robot: Planning and control for a real demonstrator," *IEEE Robotics and Automation Letters*, vol. 5, pp. 859–866, 01 2020.
- [15] Amazon FAR, P. Abbeel, J. Chen, R. Duan, A. Escontrela, M. Gandhi, S. Gundry, X. Huang, A. Kanazawa, T. Lewicki, J. Li, K. Liu, C. Rosenthal, Y. Seo, C. Sferazza, G. Shi, L. Shih, J. Tseng, Z. Wu, L. Yang, B. Yi, and Y. Zhang, "Holosoma."
- [16] C. Zheng, K. Chen, Z. Bi, Y. Li, L. Pan, J. Zhou, H. Li, and J. Ma, "Embracing bulky objects with humanoid robots: Whole-body manipulation with reinforcement learning," 2025. [Online]. Available: <https://arxiv.org/abs/2509.13534>
- [17] J. Z. Zhang, M. Sorokin, J. Brüdigam, B. Hung, S. Phillips, D. Yershov, F. Niroui, T. Zhao, L. Feroselle, X. Zhu, C. Cao, D. Ta, T. Pang, J. Wang, P. Culbertson, Z. Manchester, and S. Le Cléac'h, "Sumo: Dynamic and generalizable whole-body loco-manipulation," *arXiv preprint arXiv:2604.08508*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.08508>
- [18] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.
- [19] I. Dadiotis, N. Tsagarakis, and M. Hutter, "Dynamic object goal pushing with mobile manipulators through model-free constrained reinforcement learning," 05 2025, pp. 13 363–13 369.
- [20] A. Miller and P. Allen, "Graspt! a versatile simulator for robotic grasping," *IEEE Robotics and Automation Magazine*, vol. 11, no. 4, pp. 110–122, 2004.
- [21] S. Kumra, S. Joshi, and F. Sahin, "Antipodal robotic grasping using generative residual convolutional neural network," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 9626–9633.
- [22] M. A. Roa and R. Suárez, "Grasp quality measures: review and performance," *Auton. Robots*, vol. 38, no. 1, p. 65–88, Jan. 2015. [Online]. Available: <https://doi.org/10.1007/s10514-014-9402-3>
- [23] H.-S. Fang, C. Wang, H. Fang, M. Gou, J. Liu, H. Yan, W. Liu, Y. Xie, and C. Lu, "Anygrasp: Robust and efficient grasp perception in spatial and temporal domains," *IEEE Transactions on Robotics (T-RO)*, 2023.
- [24] M. Sundermeyer, A. Mousavian, R. Triebel, and D. Fox, "Contact-graspnet: Efficient 6-dof grasp generation in cluttered scenes," 2021.
- [25] J. Mahler, M. Matl, X. Liu, A. Li, D. Gealy, and K. Goldberg, "Dex-net 3.0: Computing robust robot suction grasp targets in point clouds using a new analytic model and deep learning," *arXiv preprint arXiv:1709.06670*, 2017.
- [26] M. B. Horowitz and J. W. Burdick, "Combined grasp and manipulation planning as a trajectory optimization problem," in *2012 IEEE International Conference on Robotics and Automation*, 2012, pp. 584–591.
- [27] Z. Gao, J. Deng, Z. Wan, H. Zhang, Y. Wang, and J. Hu, "Grasp-clip: Pick up what you want," in *2025 IEEE International Conference on Industrial Technology (ICIT)*, 2025, pp. 1–6.
- [28] H. Wang, Y. Wen, and Y. Zhang, "Tgdf: Task-oriented grasping through dense detection and foundation models," *Pattern Recognition Letters*, vol. 204, pp. 15–20, 2026. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0167865526001017>
- [29] F. Kaynar, S. Rajagopalan, S. Zhou, and E. Steinbach, "Remote task-oriented grasp area teaching by non-experts through interactive segmentation and few-shot learning," 2023. [Online]. Available: <https://arxiv.org/abs/2303.10195>
- [30] K. Black, N. Brown, J. Darpanian, K. Dhabalia, D. Driess, A. Esmail, M. R. Equi, C. Finn, N. Fusai, M. Y. Galliker, D. Ghosh, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, D. LeBlanc, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, A. Z. Ren, L. X. Shi, L. Smith, J. T. Springenberg, K. Stachowicz, J. Tanner, Q. Vuong, H. Walke, A. Walling, H. Wang, L. Yu, and U. Zhilinsky, " $\pi_{0.5}$: a vision-language-action model with open-world generalization," in *Proceedings of The 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, J. Lim, S. Song, and H.-W. Park, Eds., vol. 305. PMLR, 27–30 Sep 2025, pp. 17–40. [Online]. Available: <https://proceedings.mlr.press/v305/black25a.html>
- [31] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, S. Jakubczak, T. Jones, L. Ke, S. Levine, A. Li-Bell, M. Mothukuri, S. Nair, K. Pertsch, L. X. Shi, J. Tanner, Q. Vuong, A. Walling, H. Wang, and U. Zhilinsky, " π_0 : A vision-language-action flow model for general robot control," 2026. [Online]. Available: <https://arxiv.org/abs/2410.24164>
- [32] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog, J. Hsu, J. Ibarz, B. Ichter, A. Irpan, T. Jackson, S. Jesmonth, N. J. Joshi, R. C. Julian, D. Kalashnikov, Y. Kuang, I. Leal, K.-H. Lee, S. Levine, Y. Lu, U. Malla, D. Manjunath, I. Mordatch, O. Nachum, C. Parada, J. Peralta, E. Perez, K. Pertsch, K. Rao, K. Reymann, M. S. Ryoo, G. Salazar, P. R. Sanketi, K. Sayed, J. Singh, S. A. Sontakke, A. Stone, C. Tan, H. Tran, V. Vanhoucke, S. B. V. Vega, Q. H. Vuong, F. Xia, T. Xiao, P. Xu, S. Xu, T. Yu, and B. Zitkovich, "Rt-1: Robotics transformer for real-world control at scale," *ArXiv*, vol. abs/2212.06817, 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:254591260>
- [33] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, K. Choromanski, T. Ding, D. Driess, K. A. Dubey, C. Finn, P. R. Florence, C. Fu, M. G. Arenas, K. Gopalakrishnan, K. Han, K. Hausman, A. Herzog, J. Hsu, B. Ichter, A. Irpan, N. J. Joshi, R. C. Julian, D. Kalashnikov, Y. Kuang, I. Leal, S. Levine, H. Michalewski, I. Mordatch, K. Pertsch, K. Rao, K. Reymann, M. S. Ryoo, G. Salazar, P. R. Sanketi, P. Sermanet, J. Singh, A. Singh, R. Soricut, H. Tran, V. Vanhoucke, Q. H. Vuong, A. Wahid, S. Welker, P. Wohlhart, T. Xiao, T. Yu, and B. Zitkovich, "Rt-2: Vision-language-action models transfer web knowledge to robotic control," *ArXiv*, vol. abs/2307.15818, 2023. [Online]. Available: <https://api.semanticscholar.org/CorpusID:260293142>
- [34] Octo Model Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, C. Xu, J. Luo, T. Kreiman, Y. Tan, L. Y.

- Chen, P. Sanketi, Q. Vuong, T. Xiao, D. Sadigh, C. Finn, and S. Levine, "Octo: An open-source generalist robot policy," in *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024.
- [35] M. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, "Openvla: An open-source vision-language-action model," *arXiv preprint arXiv:2406.09246*, 2024.
- [36] T. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," *ArXiv*, vol. abs/2304.13705, 2023. [Online]. Available: <https://api.semanticscholar.org/CorpusID:258331658>
- [37] Z. Bing, Y. Yun, K. Huang, and A. Knoll, "Context-based meta-reinforcement learning with bayesian nonparametric models," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 10, pp. 6948–6965, 2024.
- [38] M. Wang, S. Tian, A. Swann, O. Shorinwa, J. Wu, and M. Schwager, "Phys2real: Fusing vlm priors with interactive online adaptation for uncertainty-aware sim-to-real manipulation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.
- [39] A. Shankar, J. Blumenkamp, and A. Prorok, "Language-conditioned offline rl for multi-robot navigation," 05 2025, pp. 14 984–14 991.
- [40] F. Ye, Y. Lin, Y. Lai, and Q. Huang, "Task-oriented language grounding for robot via learning object mask," in *2024 6th International Conference on Robotics, Intelligent Control and Artificial Intelligence (RICAI)*, 2024, pp. 16–22.
- [41] J. Hu, R. Hendrix, A. Farhadi, A. Kembhavi, R. Martin-Martin, P. Stone, K.-H. Zeng, and K. Ehsani, "Flare: Achieving masterful and adaptive robot policies with large-scale reinforcement learning fine-tuning," *arXiv*, 2024.
- [42] M. Nakamoto, O. Mees, A. Kumar, and S. Levine, "Steering your generalists: Improving robotic foundation models via value guidance," *Conference on Robot Learning (CoRL)*, 2024.
- [43] T. Schaul, D. Horgan, K. Gregor, and D. Silver, "Universal value function approximators," in *Proceedings of the 32nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, F. Bach and D. Blei, Eds., vol. 37. Lille, France: PMLR, 07–09 Jul 2015, pp. 1312–1320. [Online]. Available: <https://proceedings.mlr.press/v37/schaul15.html>
- [44] M. Andrychowicz, F. Wolski, A. Ray, J. Schneider, R. Fong, P. Welinder, B. McGrew, J. Tobin, P. Abbeel, and W. Zaremba, "Hindsight experience replay," in *Proceedings of the 31st International Conference on Neural Information Processing Systems*, ser. NIPS'17. Red Hook, NY, USA: Curran Associates Inc., 2017, p. 5055–5065.
- [45] X. Feng, "Multi-goal reinforcement learning via exploring successor matching," in *2022 IEEE Conference on Games (CoG)*, 2022, pp. 401–408.
- [46] Y. W. Teh, V. Bapst, W. M. Czarnecki, J. Quan, J. Kirkpatrick, R. Hadsell, N. Heess, and R. Pascanu, "Distral: robust multitask reinforcement learning," in *Proceedings of the 31st International Conference on Neural Information Processing Systems*, ser. NIPS'17. Red Hook, NY, USA: Curran Associates Inc., 2017, p. 4499–4509.
- [47] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [48] E. Xiang, H. Ma, X. Ma, Z. Liu, and D. Huang, "Graspldp: Towards generalizable grasping policy via latent diffusion," *arXiv preprint arXiv:2602.22862*, 2026.
- [49] B. Hu, H. Tian, D. Wang, H. Huang, X. Zhu, R. Walters, and R. Platt, "Push-grasp policy learning using equivariant models and grasp score optimization," *IEEE Robotics and Automation Letters*, 2025.
- [50] E. Foong, Y. Li, H. Jiang, G. S. Sukhatme, and D. Seita, "Handful: Sequential grasp-conditioned dexterous manipulation with resource awareness," 2026. [Online]. Available: <https://arxiv.org/abs/2604.25126>
- [51] A. Kumar, Z. Fu, D. Pathak, and J. Malik, "RMA: Rapid motor adaptation for legged robots," *Robotics: Science and Systems*, 2021.
- [52] C. Schwarke, M. Mittal, N. Rudin, D. Hoeller, and M. Hutter, "Rsl-rl: A learning library for robotics research," *arXiv preprint arXiv:2509.10771*, 2025.
- [53] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrugg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg, R. Gasoto, L. Gulich, Y. Guo, M. Gussert, A. Hansen, M. Kulkarni, C. Li, W. Liu, V. Makoviychuk, G. Malczyk, H. Mazhar, M. Moghani, A. Murali, M. Noseworthy, A. Poddubny, N. Ratliff, W. Rehberg, C. Schwarke, R. Singh, J. L. Smith, B. Tang, R. Thaker, M. Trepte, K. V. Wyk, F. Yu, A. Millane, V. Ramasamy, R. Steiner, S. Subramanian, C. Volk, C. Chen, N. Jawale, A. V. Kuruttukulam, M. A. Lin, A. Mandekar, K. Patzwaldt, J. Welsh, H. Zhao, F. Anes, J.-F. Laffleche, N. Moënné-Loccoz, S. Park, R. Stepinski, D. V. Gelder, C. Amevor, J. Carius, J. Chang, A. H. Chen, P. de Heras Ciecchowski, G. Daviet, M. Mohajerani, J. von Mural, V. Reutsky, M. Sauter, S. Schirm, E. L. Shi, P. Terdiman, K. Vilella, T. Widmer, G. Yeoman, T. Chen, S. Grizan, C. Li, L. Li, C. Smith, R. Wiltz, K. Alexis, Y. Chang, D. Chu, L. J. Fan, F. Farshidian, A. Handa, S. Huang, M. Hutter, Y. Narang, S. Pouya, S. Sheng, Y. Zhu, M. Macklin, A. Moravanszky, P. Reist, Y. Guo, D. Hoeller, and G. State, "Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning," *arXiv preprint arXiv:2511.04831*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.04831>
- [54] Epic Games, Inc., "Fab," <https://www.fab.com/>, 2026, 3D asset marketplace.
- [55] S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 13, no. 4, pp. 376–380, 1991.